DESIGN TASK

B. Akshaya

192424336

1. A healthcare organization has collected patient waiting-time data from 20 hospitals over one year.

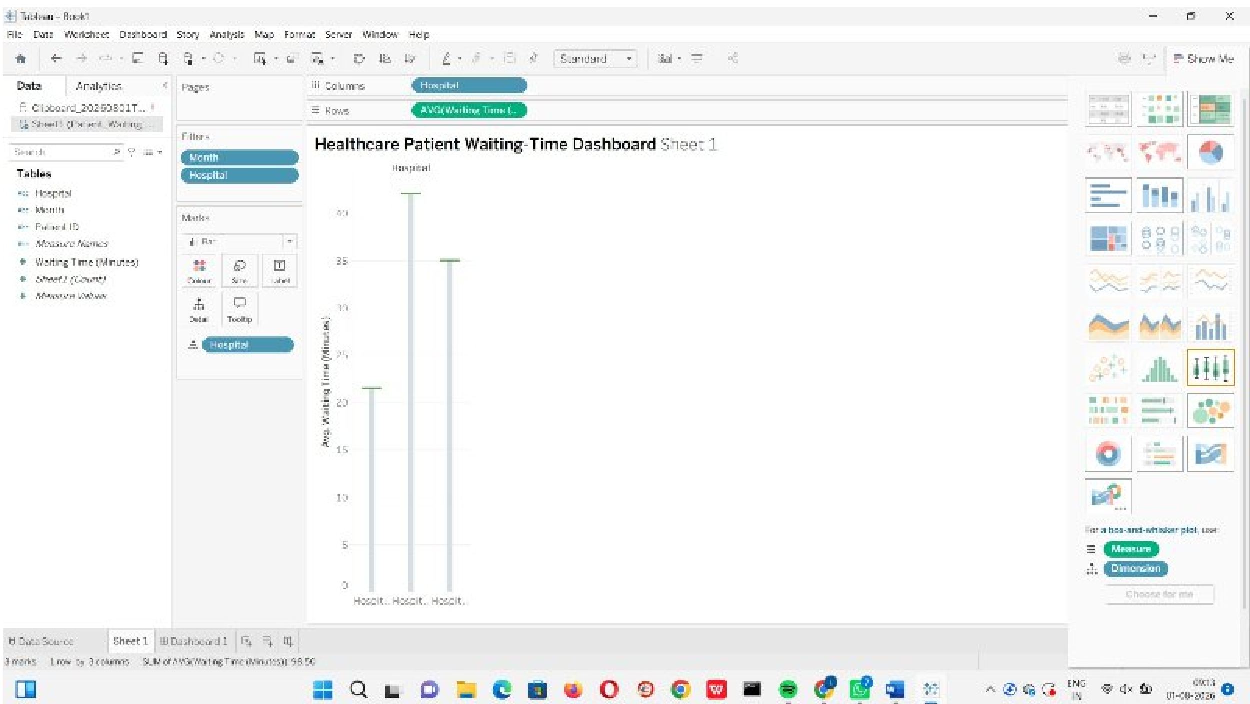
Task: Design a visualization dashboard that:

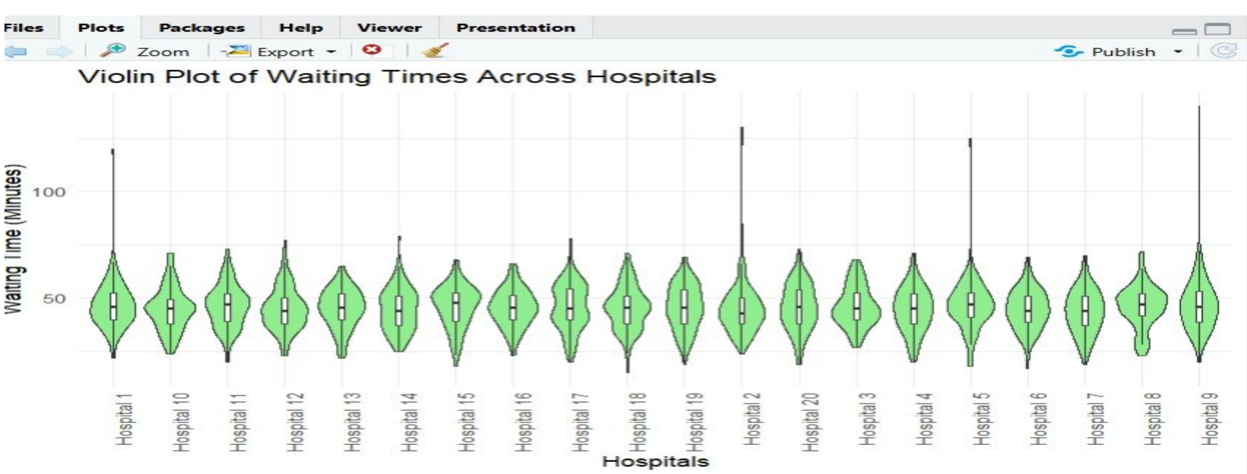
Shows Box Plots and Violin Plots for waiting times across hospitals.

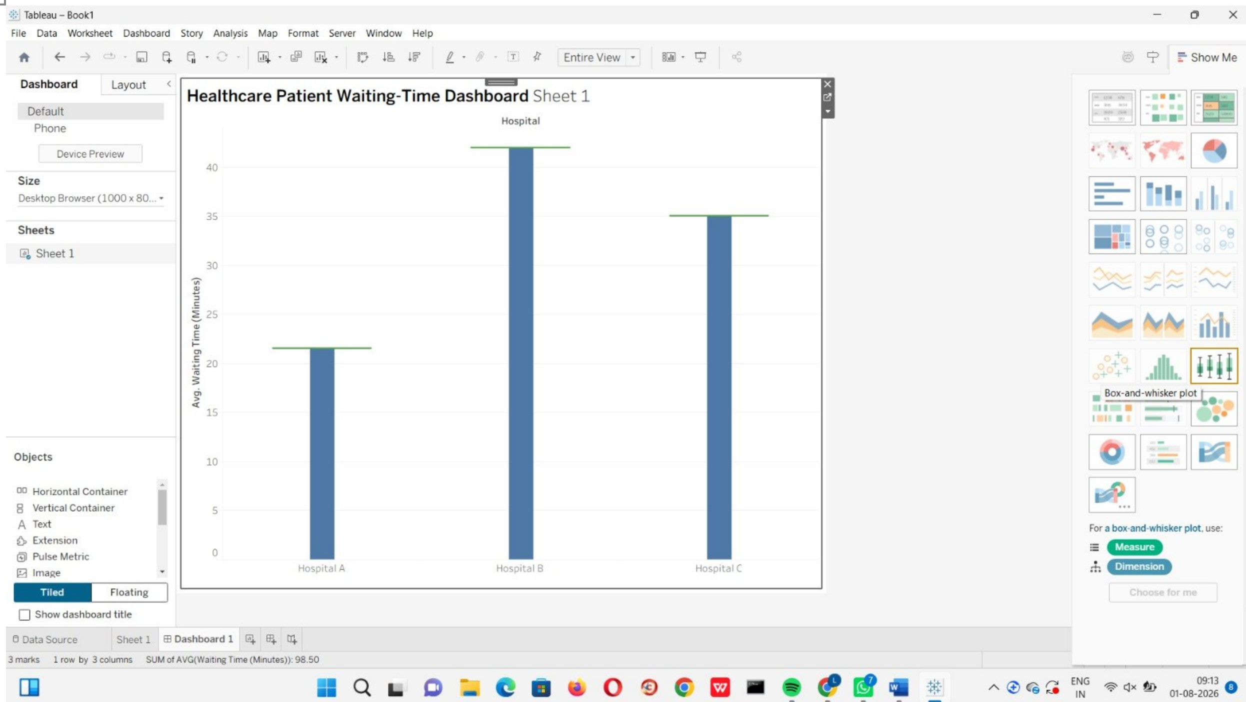
Identifies outliers and extreme waiting-time cases.

Highlights differences in median waiting times and variability among hospitals.

Objective: Help hospital administrators identify service inefficiencies and improve patient care quality.

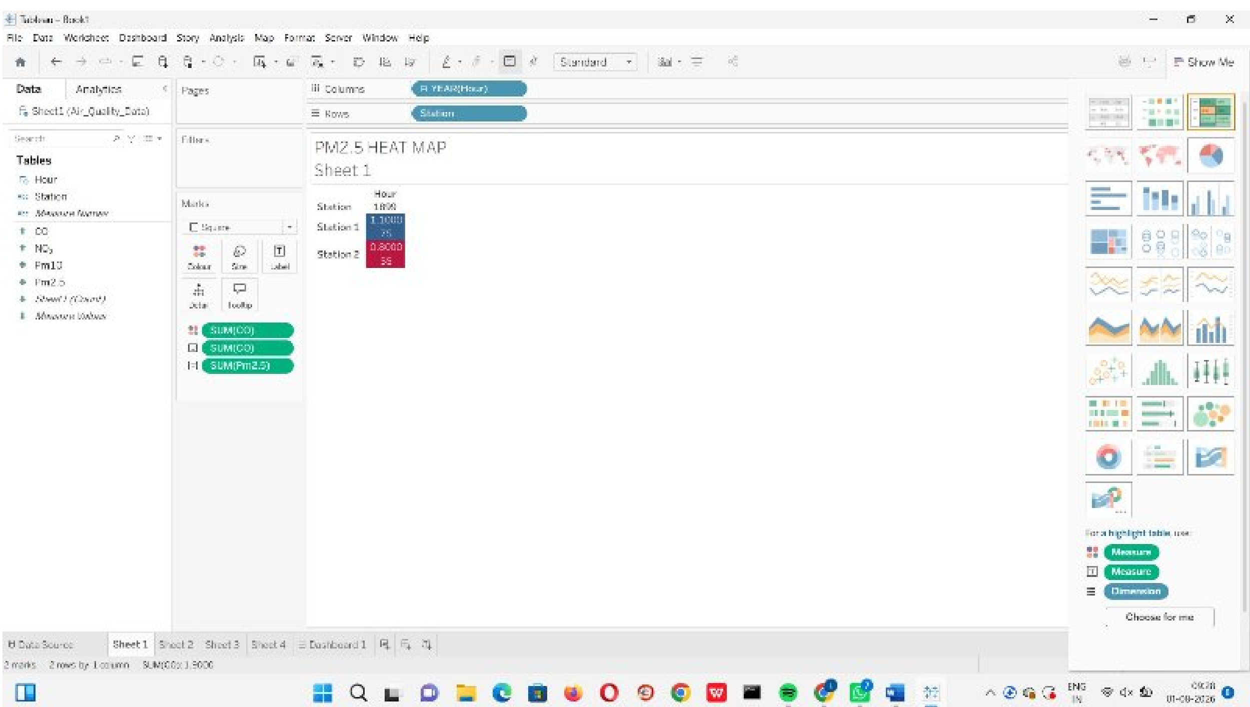






2. A smart city project monitors hourly air quality data (PM2.5, PM10, NO_2, and CO) from 15 monitoring stations.
Task: Create a visualization design that:
Uses Heatmaps to display pollution intensity across locations and time periods.
Employs Ridgeline Plots to compare pollutant distributions among stations.
Highlights pollution spikes and abnormal environmental conditions.

Objective: Assist environmental authorities in identifying pollution hotspots and planning mitigation measures.



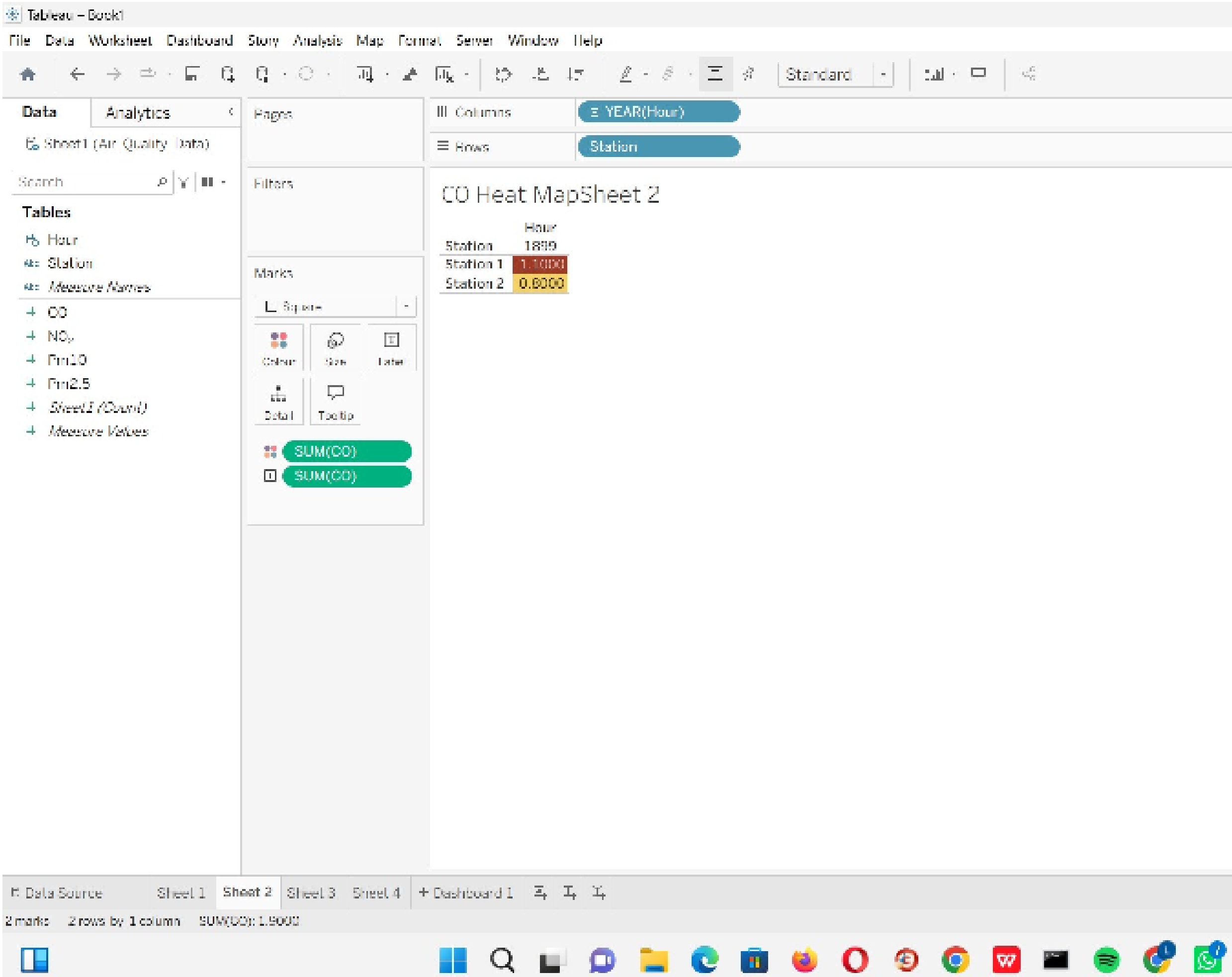


Tableau - Book1
File Data Worksheet Dashboard Story Analysis Map Format Server Window Help
Data Analytics
Pages
Columns YEAR(Hour)
Rows Station
CO Heat MapSheet 2
Station 1 1.1000
Station 2 0.8000
SUM(CO)
Data Source Sheet 1 Sheet 2 Sheet 3 Sheet 4 Dashboard 1
2 marks 2 rows by 1 column SUM(CO): 1.9000

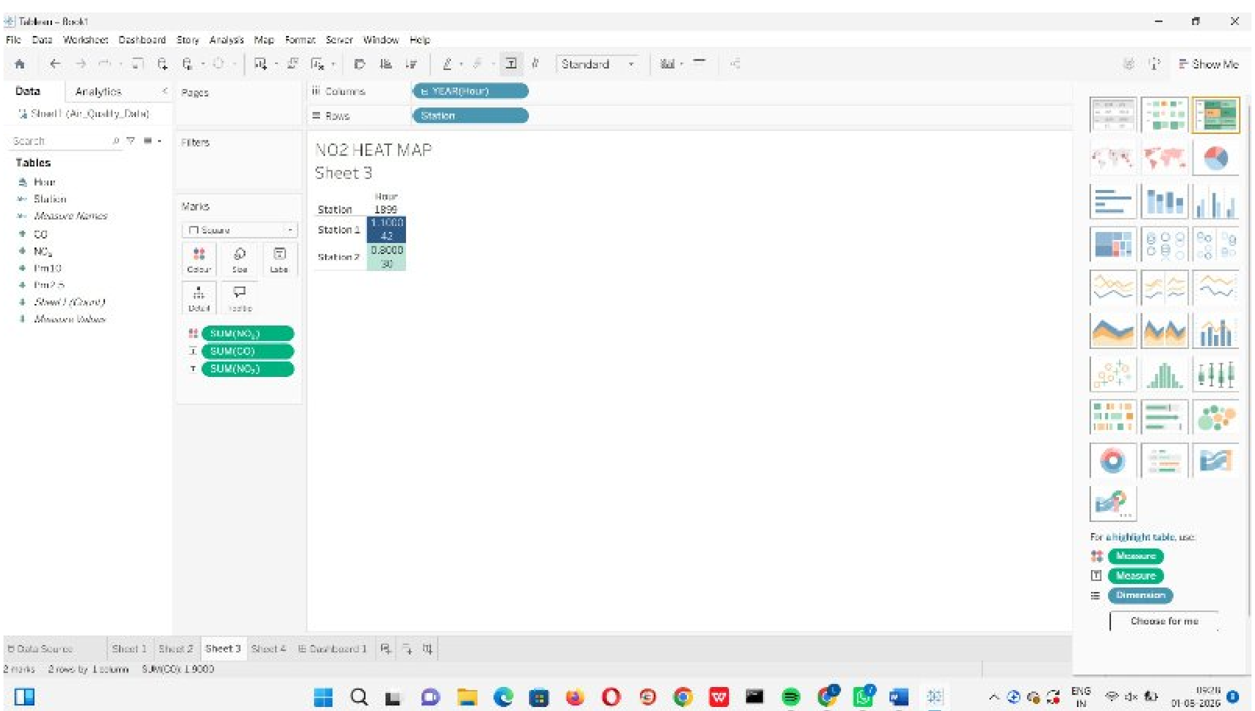
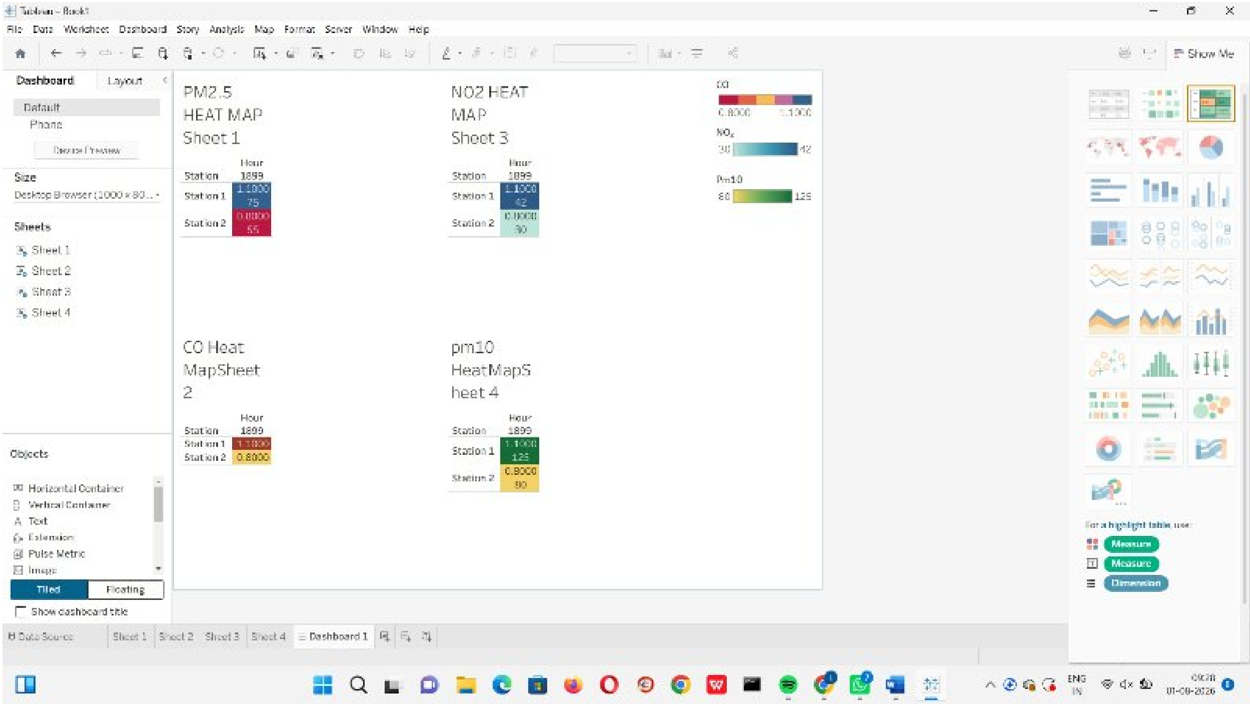


Tableau - Book1
NO2 HEAT MAP
Sheet 3
Station 1 1.1000 42
Station 2 0.8000 30
Show Me


PM2.5
HEAT MAP
Sheet 1
NO2 HEAT
MAP
Sheet 3
CO Heat
MapSheet
2
pm10
HeatMapS
heet 4
Station
Hour
1899
Station 1
Station 2
Dashboard
Layout
Default
Phone
Size
Sheets
Sheet 1
Sheet 2
Sheet 3
Sheet 4
Objects
Horizontal Container
Vertical Container
Text
Extension
Pulse Metric
Image
Tiled
Floating
Show dashboard title
Show Me
Data Source
Dashboard 1

3. A university tracks student academic performance across multiple departments over five academic years.

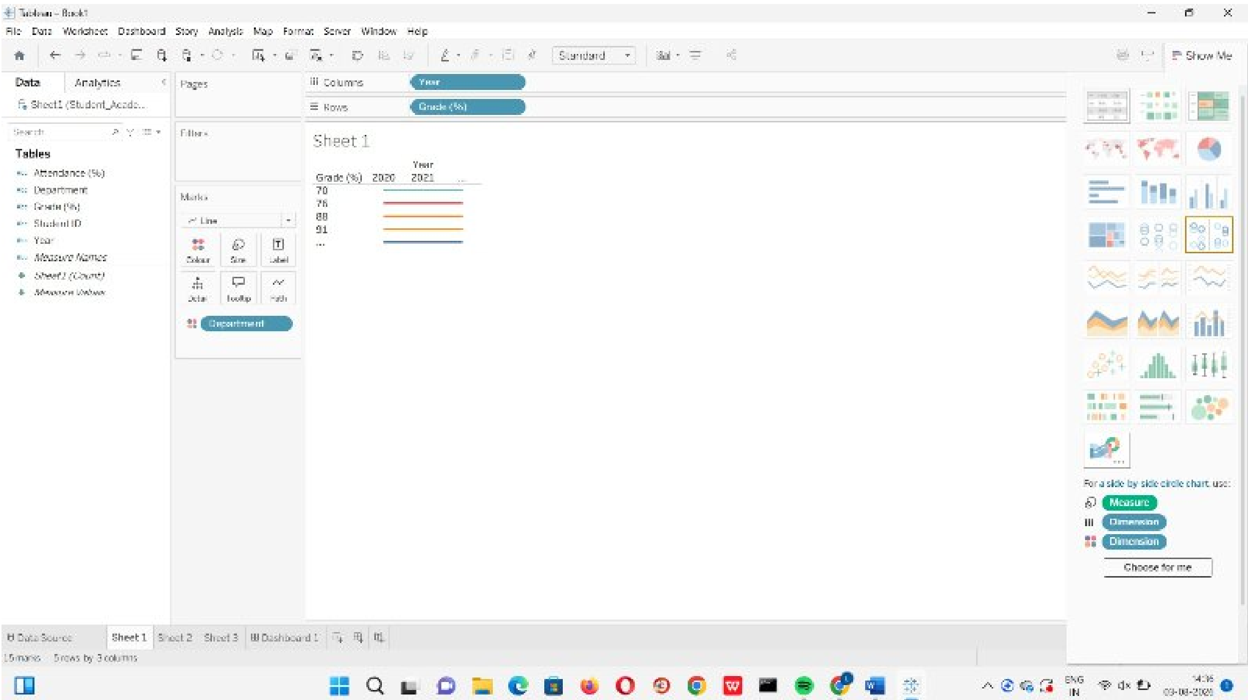
Task: Design a visualization dashboard that:

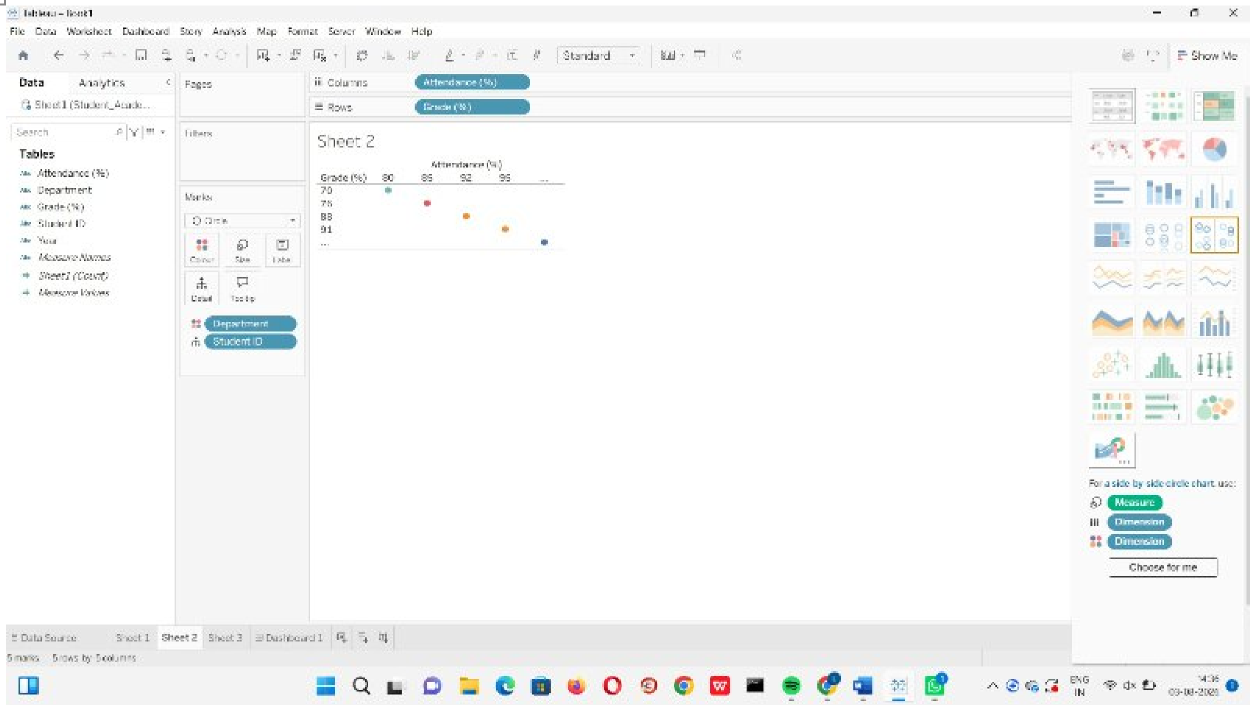
Uses Line Charts to show performance trends over time.

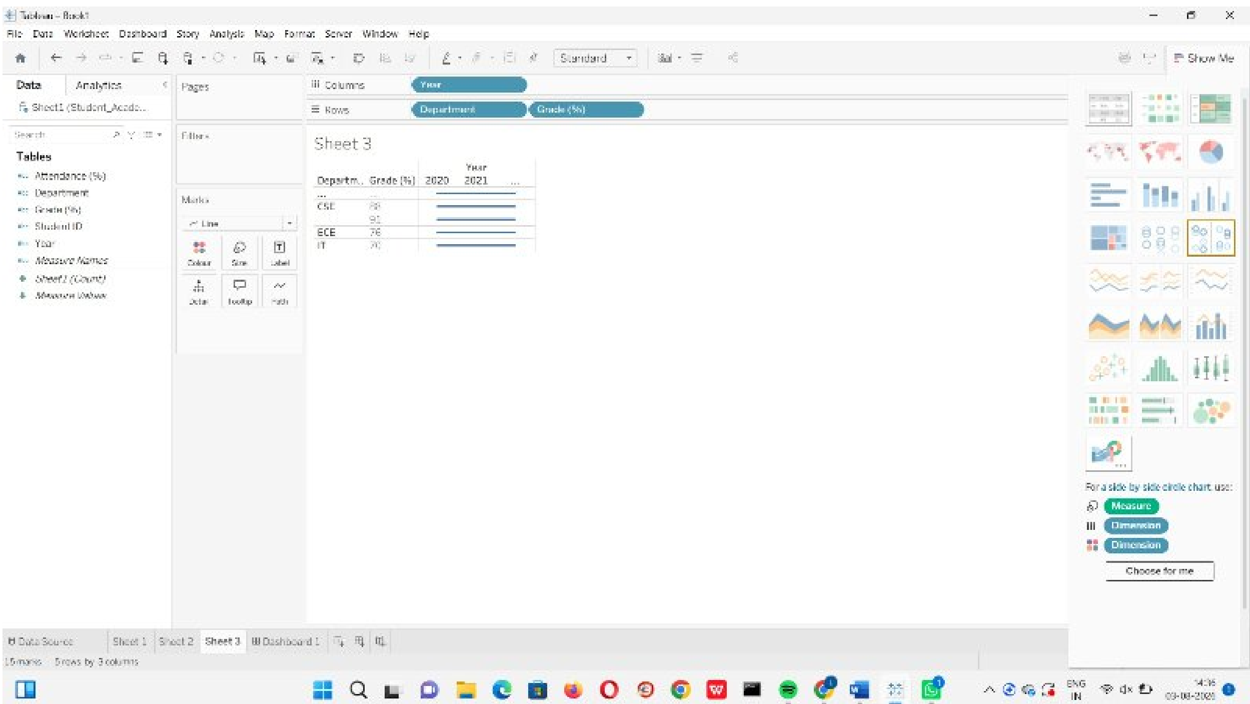
Incorporates Scatter Plots to analyze the relationship between attendance and grades.

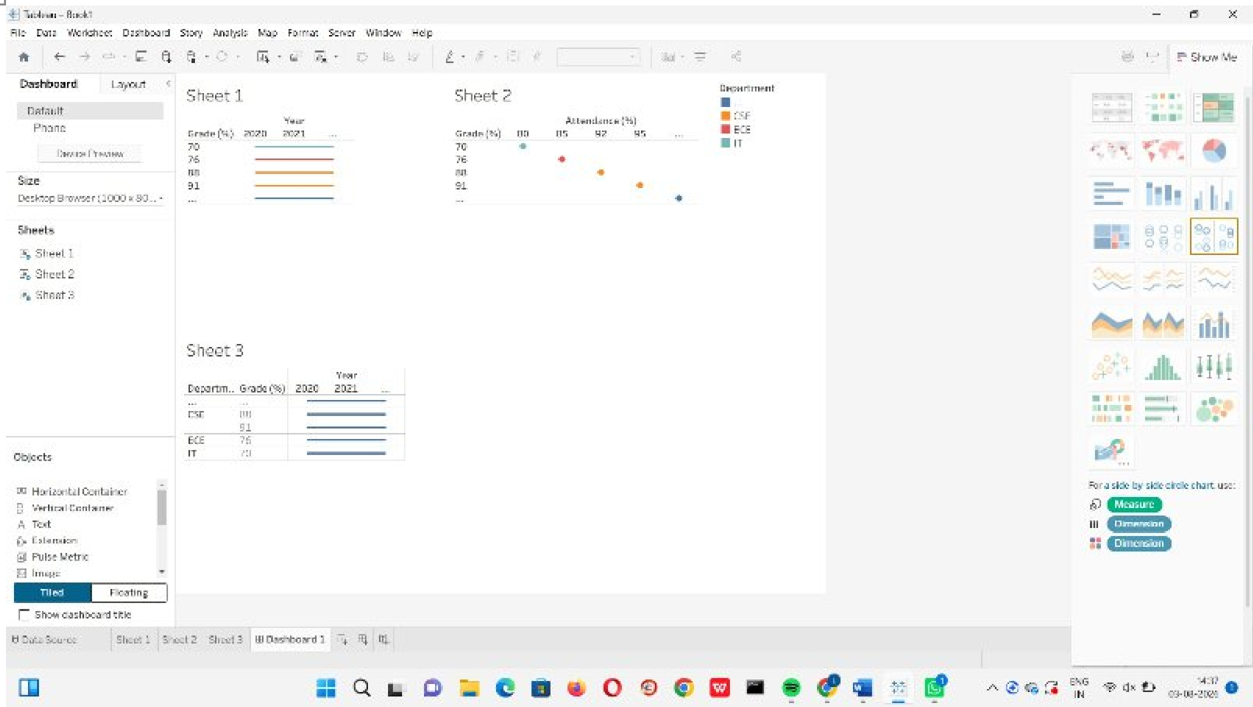
Uses Facet Grids to compare departmental performance simultaneously.

Objective: Enable academic administrators to identify performance trends and factors affecting student success.



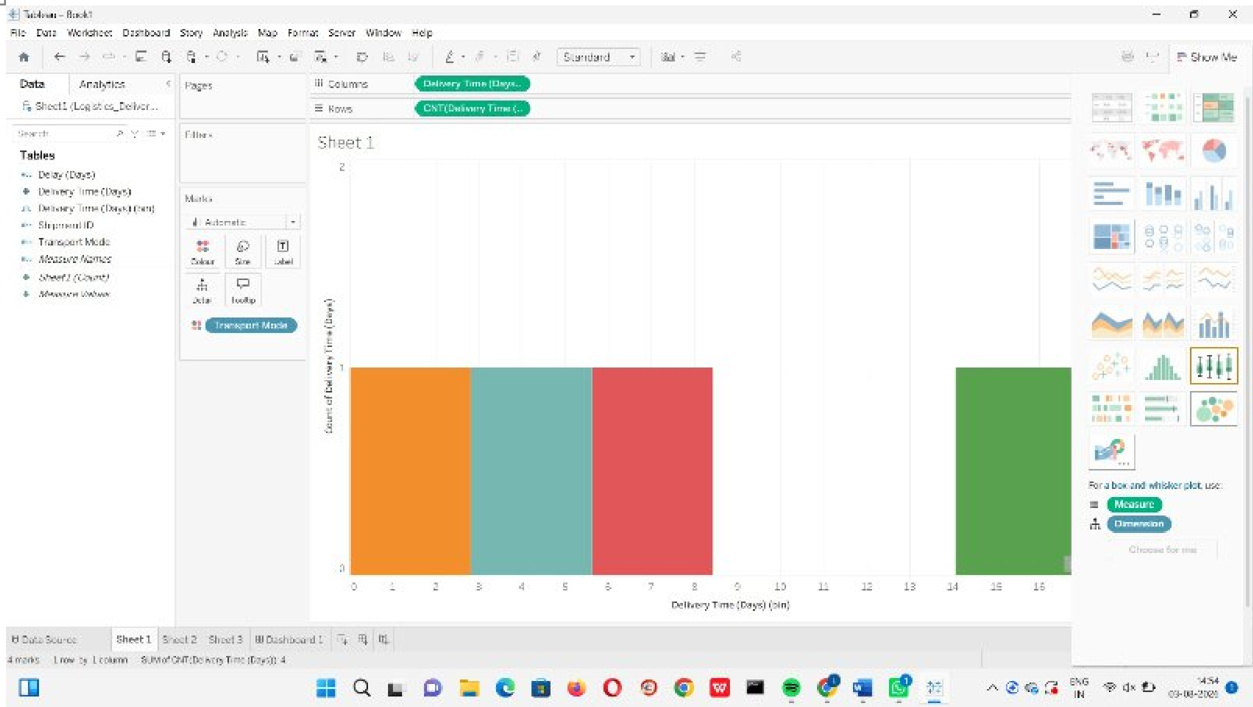


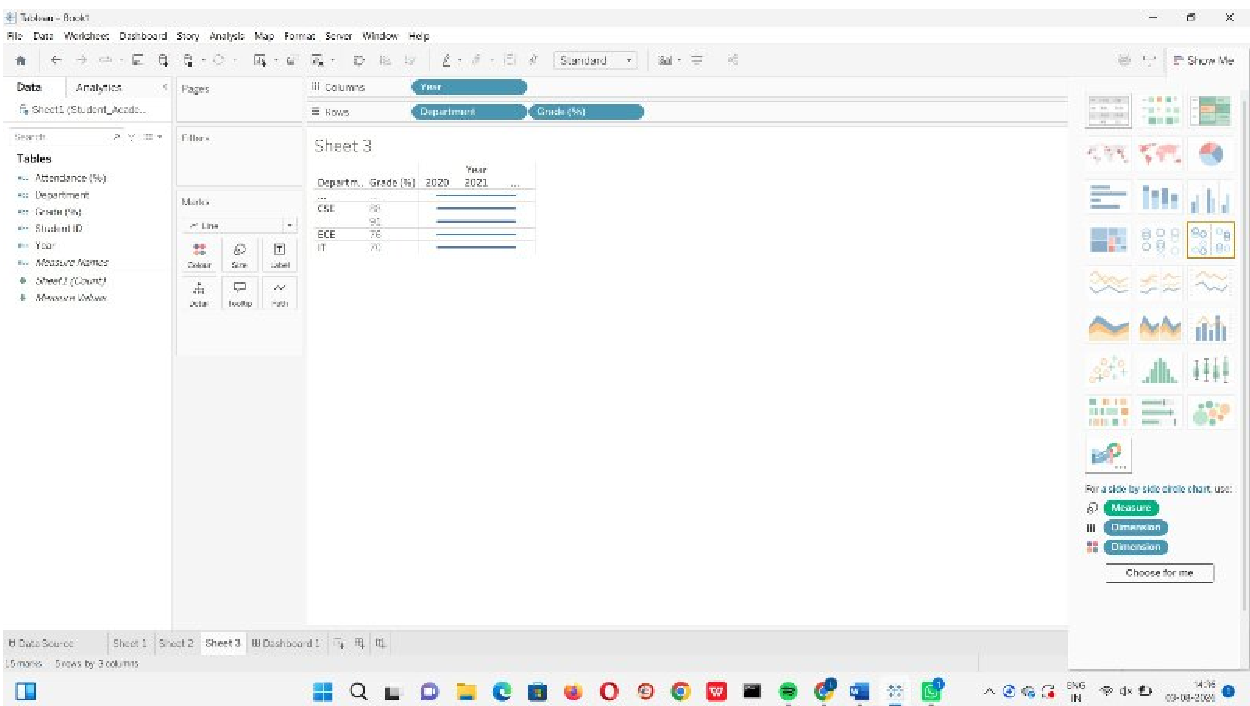


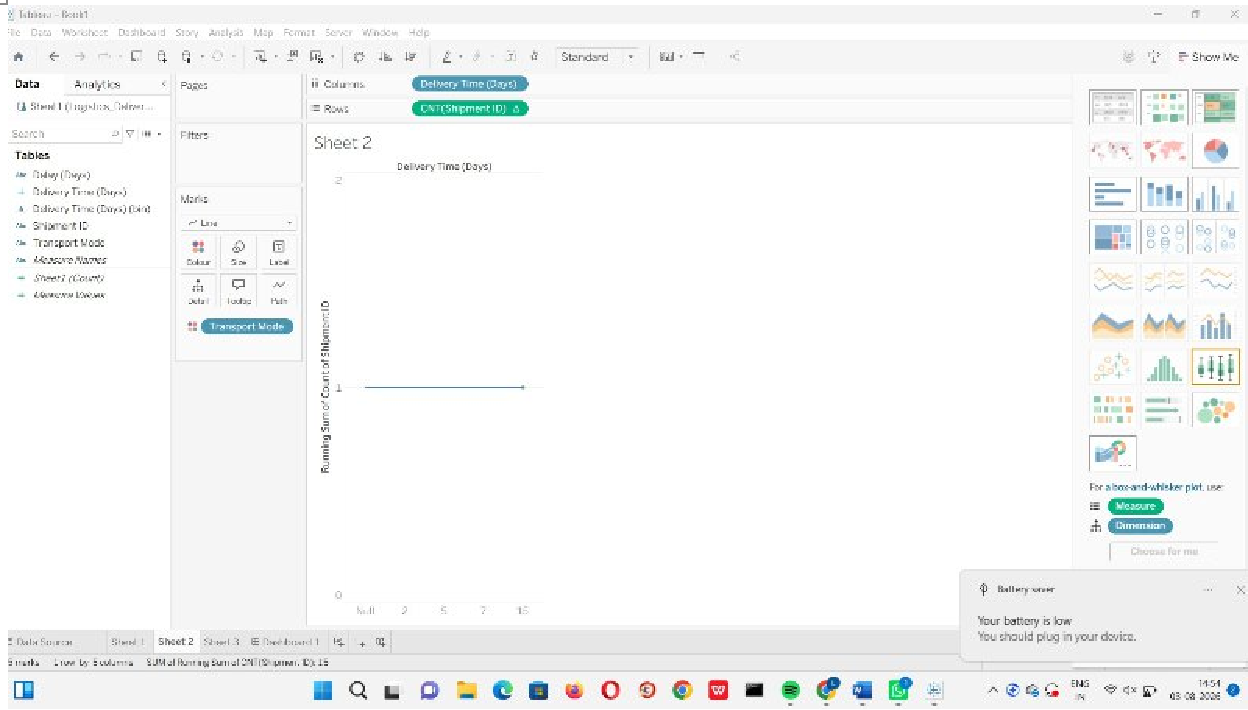


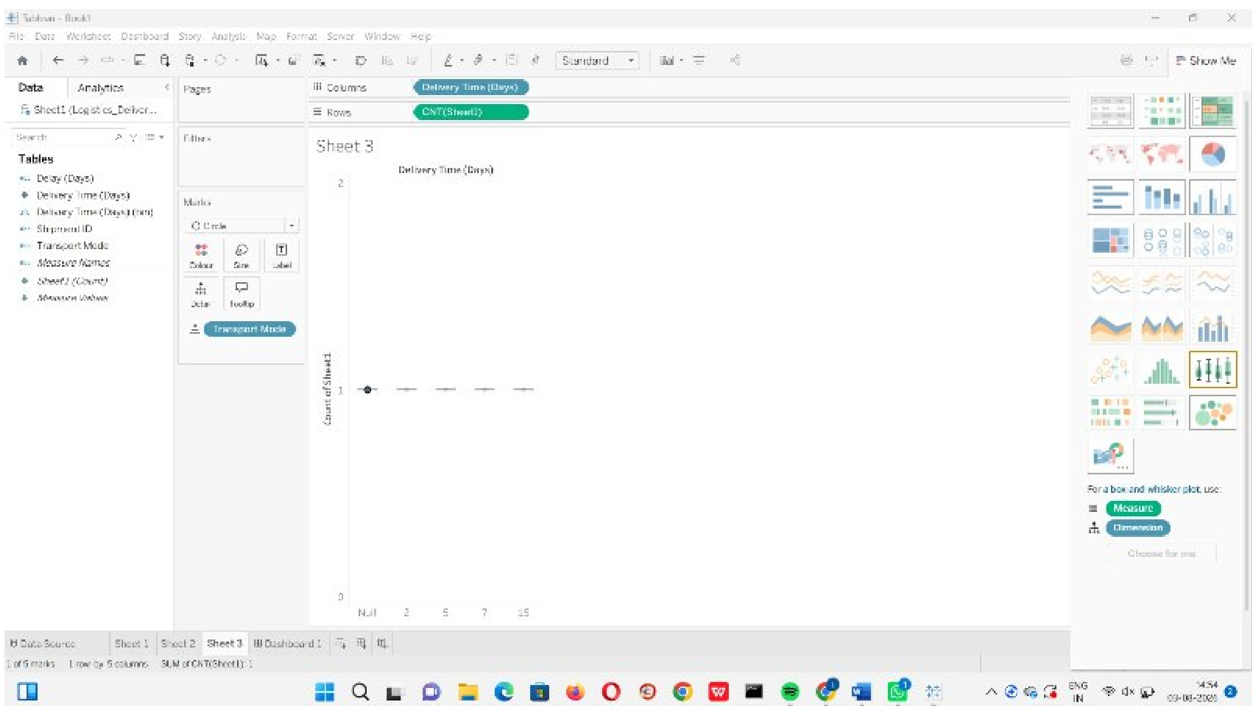
4. A logistics company records delivery times for thousands of shipments using Road, Rail, Air, and Sea transportation modes.
Task: Create a visualization design that:
Uses Histograms and ECDFs to compare delivery-time distributions across transportation modes.
Employs Q-Q Plots to assess whether delivery times follow a theoretical normal distribution.
Highlights delays, variability, and outliers in delivery performance.

Objective: Support operational teams in optimizing transportation strategies and improving delivery reliability.


Sheet 1
Columns: Delivery Time (Days...
Rows: CNT(Delivery Time (...
Count of Delivery Time (Days)
Delivery Time (Days) (bin)
For a box and whisker plot, use:


Sheet 3
Columns: Year
Rows: Department, Grade (%)
Year
Departm.. Grade (%) 2020 2021
CSE
ECE
IT
For a side by side circle chart use:





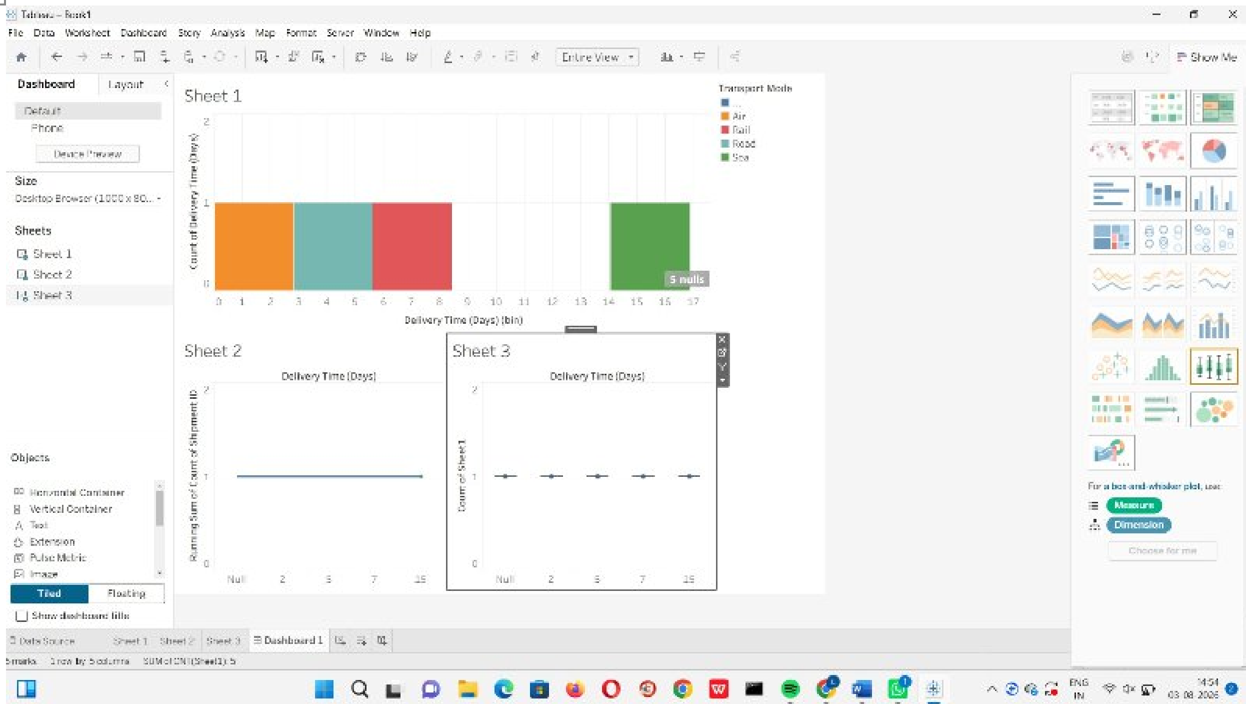


Tableau - Book1
File Data Worksheet Dashboard Story Analysis Map Format Server Window Help
Entire View
Show Me
Dashboard
Layout
Default
Phone
Device Preview
Size
Desktop Browser (1000 x 80...
Sheets
Sheet 1
Sheet 2
Sheet 3
Objects
Horizontal Container
Vertical Container
Text
Extension
Pulse Metric
Image
Tiled
Floating
Show dashboard title
Sheet 1
Transport Mode
Air
Rail
Road
Sea
Count of Delivery Time (Days)
5 nulls
Delivery Time (Days) (bin)
Sheet 2
Delivery Time (Days)
Running Sum of Count of Shipment ID
Null
Sheet 3
Delivery Time (Days)
Count of Sheet1
For a box-and-whisker plot, use
Measure
Dimension
Choose for me
Data Source
Sheet 1
Sheet 2
Sheet 3
Dashboard 1
5 marks
SUM of CNT(Sheet1): 5
ENG
IN